

GENERAL PHARMACOLOGY

MBS230-P&T

UNIT 1.1.9

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LS 1.1.9: THERAPEUTIC DRUG MONITORING

Learning Objectives

At the end of the Class, Students should be to:

- Define TDM
- Identify factors needed to consider when doing TDM
- Know why TDM is important

Introduction

TDM- Effective way of monitoring drug levels in plasma.

Purpose: TDM is employed to measure blood drug levels so that the most effective dosage can be determined, with toxicity prevented. TDM is also utilized to identify noncompliant patients (those patients who, for whatever reason, either cannot or will not comply with drug dosages as prescribed by the physician).

Things to consider

Because so many different factors influence blood drug levels, the following points should be taken into consideration during TDM:

- the age and weight of the patient
- the route of administration of the drug
- the drug's absorption rate, excretion rate, delivery rate, and dosage;
- other medications the patient is taking
- other diseases the patient has
- the patient's compliance regarding the drug treatment regimen
- the laboratory methods used to test for the drug.

Some Factors to consider

DOSAGE REGIMEN:

- It is one of the factors to be considered while interpreting TDM data. It is important to know the duration of drug therapy, dosage and when the last dosage was taken.

ACTIVE METABOLITE:

- Many drugs are biotransformed into compounds that are pharmacologically active. When evaluating the therapeutic effect of such drugs, the relative contributions of all active substances present in the serum must be integrated e.g. imipramine is biotransformed to the active metabolite – desipramine.

Some Factors to consider

EFFECT OF DISEASE STATES:

- Acute or chronic disease alters drug clearance patterns. Thus drug concentrations may be elevated or depressed depending on the pathophysiology of the system involved, e.g., liver disease impairs the clearance of drugs dependent upon conversion to more water soluble compounds. Congestive cardiac failure can precipitate elevated drug levels of agents dependent on hepatic metabolism for clearance.

FREE DRUG MONITORING:

- Development of new filtration devices (equilibrium dialysis, ultrafiltration, ultracentrifugation) has made it possible to measure free unbound drug levels in serum. The advantages are that the free concentrations is independent of changes in plasma binding and is the pharmacologically active concentration.

Some Factors to consider

USE OF SALIVA IN DRUG MONITORING:

- The concentration of a drug in saliva is proportional to the concentration of the unbound rather than to the total of bound and unbound drugs in plasma. The practice of measuring drugs in saliva is appealing because it is non invasive.

EFFECT OF AGE:

- Variability in response to drugs occurs at extremes of age. Elderly patients are more sensitive to the CNS depressant effect of drugs but are less sensitive to cardiovascular effects of propranolol. On the other hand young children are more sensitive to CNS depression effects of morphine. However more data are needed on the effects of age on pharmacokinetic and pharmacodynamics of drugs to allow optional individualization of dosage.

WHY MONITOR DRUG THERAPY?

J K Aronson, M Hardman



The peak expiratory flow rate in patients with asthma can be used to monitor diurnal variation in airways resistance and the effects of bronchodilators.

It is common clinical practice to monitor the response to treatment of a disease. Obvious examples include monitoring the treatment of hypertension by measuring a patient's blood pressure, the course of an infection by measuring body temperature, and asthma by measuring the peak expiratory flow rate. The effects of a drug are not often monitored, however, despite the fact that monitoring techniques are available. In this series we will discuss:

- The reasons for monitoring drug treatment
- The ways in which therapeutic responses to drugs can be measured
- Monitoring patient compliance
- Measuring and interpreting plasma drug concentrations as a means of monitoring drug treatment.

Why Monitor Drugs

1.TO SEE WHETHER THERE IS A
THERAPEUTIC RESPONSE

2.TO ASSESS DRUG TOXICITY

3.TO ASSESS COMPLIANCE

The aim of all drug treatment is to improve the patient's condition, and this aim has two parts: to achieve the maximum therapeutic benefit, and to minimise unwanted effects (such as adverse drug reactions and interactions).

Direct or indirect monitoring of drug treatment may allow the doctor to assess the degree of therapeutic response. No response might indicate that the dosage is inadequate, another drug would be better, the diagnosis is incorrect, or the patient has not taken the drug in the manner prescribed.

Monitoring drug treatment is also important to detect adverse effects. In many cases the correct dose of a drug can be found only by initial trial, subsequent monitoring, and consequent adjustment of the dosage regimen. In making adjustments it is important to remember that the dosage regimen includes the dose of the drug, the route and frequency of administration, and the duration of treatment.

Why Monitor Drugs

Examples of easily measurable therapeutic responses

- Urine output in patients treated with desmopressin for diabetes insipidus
- Intraocular Pressure in patients treated with timolol eye drops
- for glaucoma
- Muscle fatigue in patients treated with pyridostigmine for myasthenia gravis

Is there a directly measurable therapeutic response?

Sometimes the direct therapeutic response can be measured easily. In other instances, however, the interpretation of a measurable response is not so clear cut. For example, an apparent response may be due to spontaneous resolution of the disease, as when a viral sore throat improves spontaneously after an antibiotic has been prescribed inappropriately. An absent measurable therapeutic response to a drug, however, usually suggests that either a change in dosage regimen or a different therapeutic approach altogether is required.

Monitoring drug treatment

1. Want our treatments to work
 2. Do not wish treatments to cause harm.
- Monitoring drug treatment is one way of seeing that a treatment works, while protecting the patient from adverse drug effects.
 - For many patients and many treatments clinical evaluation is sufficient (e.g. Measuring blood pressure in a patient on antihypertensive treatment).
 - When therapeutic goals cannot always be directly observed, monitoring may require blood tests to determine whether therapeutic levels have been reached.e.g the measurement of the international normalised ratio (INR) in patients treated with warfarin.
 - Ensure that the therapeutic goal (prevention of thrombosis is met)measuring INR helps to avoid the risk of haemorrhage, which rises steeply as the INR increases above 2.0.

Factors to take into account when monitoring for an adverse drug effect

- **The adverse effect**

The effect should be potentially serious; The relation between the latent and overt effects should be known

- **The monitoring test**

The test should be safe, simple, precise, and validated. The distribution of test values in the exposed population should be known and suitable cut-off values established; The test should be acceptable to treated patients; A strategy in the face of a positive monitoring test should be agreed

- **The response to positive tests**

An effective intervention should exist; This early intervention should make the outcome better than it would have been with delayed intervention; Evidence for the intervention should be robust

- **The monitoring strategy**

The strategy should reduce morbidity or mortality from the adverse effect; The strategy should be acceptable to patients and professionals; Benefits of monitoring should outweigh the physical and psychological harm; The cost of monitoring should be proportionate; A system for assuring the standards of the monitoring programme should exist; Possibility of reducing or removing risks of adverse effects by selection of drug or dosage, or by pretreatment detection of susceptible people, should have been fully explored

Specific individual circumstances for TDM

1. Low therapeutic index
2. Poorly defined clinical end point
3. Non compliance
4. Therapeutic failure
5. Drugs with saturable metabolism
6. Wide variation in the metabolism of drugs
7. Major organ failure
8. Prevention of adverse drug effects

Examples of Drugs for TDM

1. Cardio active drugs : amiodarone, digoxin, digitoxin, disopyramide, lignocaine, procainamide, propranolol and quinidine
2. Antibiotics : gentamycin, amikacin and tobramycin
3. Antidepressants : lithium and tricyclic antidepressants
4. Antiepileptic drugs : Phenytoin, phenobarbitone, benzodiazepines, carbamazepine, Valproic acid and ethosuximide
5. Bronchodilators : theophylline
6. Cancer chemotherapy : methotrexate
7. Immunosuppressives : cyclosporine

CLINICAL USEFULNESS OF TDM:

- TDM data provides the clinician with greater insight into the factors determining the patients response to drug therapy. For example when a patient fails to respond to a usual therapeutic dose, measurement of plasma level can help to distinguish a non-compliant patient and a patient who is a true non-responder. (Identify therapeutic Failure).
- TDM also provides useful information regarding individual variations in drug utilization patterns and alteration in drug utilization as a consequence of altered physiological state or disease process.
- TDM is a useful adjunct in treating many patients provided the potential pit falls and problems are considered.

CLINICAL USEFULNESS OF TDM:

- TDM is a practical tool that can help the physician provide effective and safe drug therapy in patients who need medication. Maximizes Efficacy and avoids Toxicity.
- Monitoring can be used to confirm a blood drug concentration level that is above or below the therapeutic range, or if the desired therapeutic effect of the drug is not as expected. If this is the case, and dosages beyond normal then have to be prescribed, TDM can minimize the time that elapses.
- TDM is important for patients who have other diseases that can affect drug levels Or who take other medicines that may affect drug levels by interacting with drug being tested.

Examples

- Without drug monitoring, the physician cannot be sure if a patient's lack of response to an antibiotic reflects bacterial resistance, or is the result of failure to reach the proper therapeutic range of antibiotic concentration in the blood.
- In cases of life-threatening infections, timing of effective antibiotic therapy is critical to success. It is equally crucial to avoid toxicity in a seriously ill patient. Therefore, if toxic symptoms appear with standard dosages, TDM can be used to determine changes in dosing. Blood demonstrates drug action at any specific time. drug levels examined from Urine reflect the presence of a drug over many days (depending on the rate of excretion).

At the end of the Lecture, Students should be able to comprehend;

- Defination of TDM
- Listing all the Factors necessary for TDM
- Knowing the reasons for perfoming TDM and why it is important in Clinical practice